SNAPDRAGON® AI LAB / BUILD & PRESENT CHALLENGE

SOLUTION SUBMISSION ROUND

CodeSync-AI

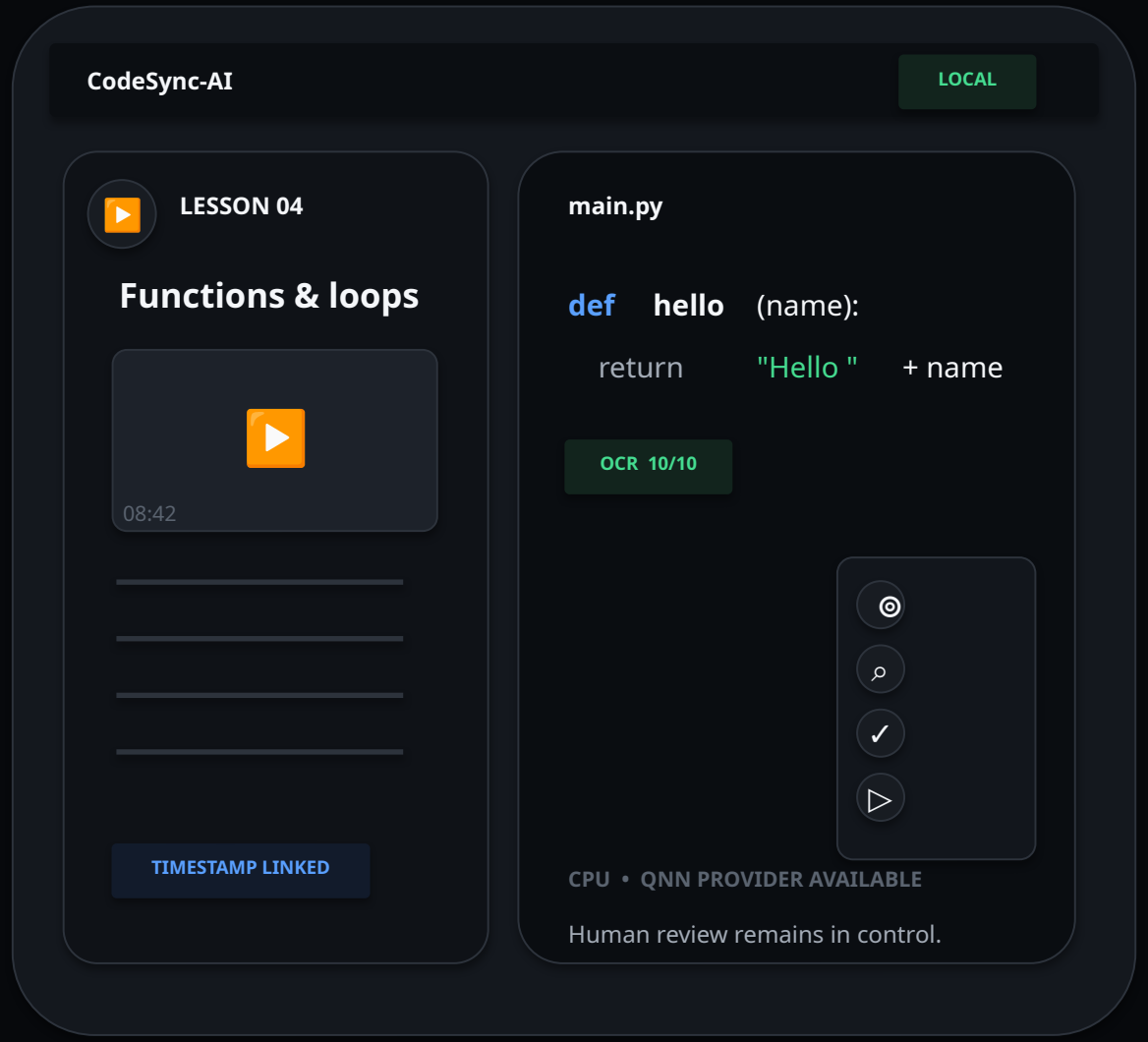
One Screen. Zero Alt-Tabs.

An offline coding classroom that preserves lesson context from instruction → code → practice → execution.

OFFLINE-FIRST | LOCAL OCR | QNN PATH

The product story

Keep the learner in the learning surface.



The hidden cost is context reconstruction.

Consulting lens: the workflow is fragmented even when every individual tool works.



THE PRODUCT BET

Preserve the learner’s place.

The lesson, captured code, timestamp and practice state should remain connected.

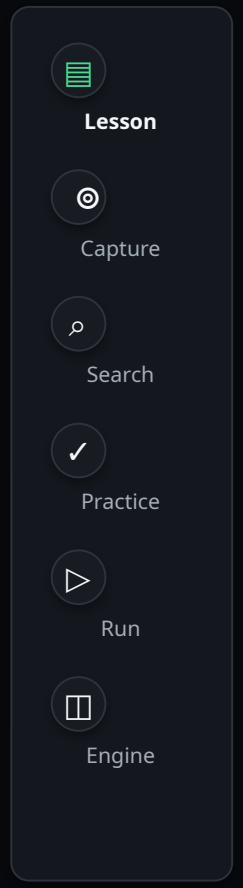
REPO EVIDENCE

- ✓ 10/10 sample code lines captured exactly
- ✓ Headless tests reported for core logic
- ✓ Offline/local workflow is implemented

DECISION QUESTION **Can the product reduce the number of times a learner must reconstruct context?**

One screen. Six connected states.

The information architecture makes the workflow visible: lesson context is the spine; AI services attach to it.



INFORMATION ARCHITECTURE

LESSON CONTEXT
video • slides • transcript • timestamp

CAPTURED CODE	OCR output • cleaned code
SEARCH INDEX	TF-IDF across lesson content
PRACTICE STATE	checkpoint • hint • pass/fail
EXECUTION	run output • error explanation

PRIMARY USER FLOW

01 **LESSON** — understand

02 **CAPTURE** — extract

03 **LINK** — trace

04 **PRACTICE** — verify

05 **RUN** — execute

06 **EXPLAIN** — recover

Context is the spine; AI is a service layer, not the destination.

From browser action to edge inference.

The technical story is simple enough to audit and concrete enough to validate on Snapdragon hardware.



RUNTIME BRANCH

CPU provider
validated baseline

QNN provider
wired path • device validation next

STATUS DISCIPLINE

VERIFIED CPU workflow

WIRED QNN

Qualcomm AI Hub supports ONNX Runtime / QNN targeting and device profiling; the submission's next step is to run this project's models through that measured path.

Show the product working, then show the measurement.

A reviewer should be able to distinguish the user experience, engineering proof, and edge-AI validation plan.

01	02	03	04	05	06
OPEN	**PAUSE**	**CAPTURE**	**TRACE**	**PRACTICE**	**RUN**
lesson	code frame	OCR	timestamp	checkpoint	explain

BUILT

- ✓ Offline lesson pack
- ✓ Local OCR capture
- ✓ Timeline linking
- ✓ Search + checkpoints
- ✓ Run + error explanation

OBSERVE

- ✓ Task time
- ✓ Context switches
- ✓ OCR correction rate
- ✓ Search relevance
- ✓ Checkpoint completion

VALIDATE NEXT

- ✓ Snapdragon NPU latency
- ✓ Memory + utilization
- ✓ Power / thermal
- ✓ Offline success

EXECUTIVE LENS — **Does the workflow reduce context reconstruction — and can the edge path prove measurable benefit?**

A focused product today. A measurable edge-AI path tomorrow.

The strongest submission story is not 'everything is finished'; it is a credible product with a disciplined validation path.

NOW

OFFLINE WORKSPACE

lesson + OCR + trace + practice + run

→

NEXT

SNAPDRAGON VALIDATION

QNN / NPU correctness + benchmark

→

THEN

MODEL OPTIMIZATION

OCR • embeddings • transcription

→

EXPAND

LOCAL ASSISTANCE

small on-device model + richer guidance

ONE SCREEN. ZERO ALT-TABS.

AI understands the code.

Edge AI keeps assistance close.

Humans remain in control.

SUBMISSION PRINCIPLE

Verified ≠ roadmap.
Measure before claiming.

Repository: github.com/kartiksaxena-max/CodeSync-AI-One-Screen-Zero-Alt-Tabs-

Qualcomm AI Hub → on-device deployment / profiling path

Repo-grounded • verified / wired / next clearly separated